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Second Cycle Degree in Quantitative Finance



Financial Risk Management

Predictive Modeling of Basis Risk - Euro STOXX 50 Futures: An Empirical Analysis

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1. Introduction

The financial derivatives market plays a pivotal role in modern risk management and price discovery. Among its key instruments, equity index futures serve as essential tools for hedging, speculation, and portfolio adjustment. A central concept in the pricing and performance of these contracts is the *basis*, defined as the difference between the futures price and the spot price of the underlying asset. While in theory the basis should reflect predictable components such as cost-of-carry, empirical evidence reveals significant deviations due to liquidity constraints, volatility, macroeconomic shocks, and market frictions. This paper investigates the determinants of basis risk in the Euro STOXX 50 equity index futures market using a multi-factor econometric framework. The Euro STOXX 50, as a representative European blue-chip index, provides a unique environment to study basis behavior given the region's diverse economic structure, fragmented sovereign risk profiles, and recent episodes of monetary tightening and geopolitical disruption.

The objective of this study is two-fold. First, we aim to quantify how economic, financial, and contract-specific variables influence the basis over time. Second, we examine whether these relationships are stable or time-varying, particularly in response to major events such as the COVID-19 pandemic and the Russia–Ukraine conflict. Understanding these dynamics is critical for effective hedging strategies, risk forecasting, and market microstructure analysis. We provide a high-frequency, five-year empirical analysis of the Euro STOXX 50 futures basis using a comprehensive dataset that integrates spot and futures prices, macroeconomic indicators, interest rate measures, liquidity proxies, and country risk variables. In doing so, the paper has implications for asset managers, market makers, and regulators concerned with pricing efficiency, basis volatility, and cross-market linkages in European financial markets.

2. Literature Review

The empirical literature on basis risk in equity index futures highlights several economic, institutional, and behavioral factors that influence the divergence between spot and futures prices. This section summarizes relevant findings across three key domains: market microstructure and liquidity, macro-financial variables, and hedging efficiency. These streams of literature provide the theoretical and empirical foundation for our model specification.

Bid-ask spreads, open interest, and speculative positioning are central to understanding short-run fluctuations in basis. Chang and Lin (2015) find that wider bid-ask spreads and lower market depth significantly increase the basis spread in Taiwanese index futures, particularly during periods of low liquidity. Similarly, Figlewski (1984) argue that open interest serves as a proxy for speculative activity, which may either amplify or dampen basis risk depending on hedger versus speculator dominance. These findings underscore the

importance of including liquidity proxies in our regression model. Market volatility, typically captured by indices such as VIX or VSTOXX, affects basis risk by altering investor risk appetite and demand for hedging. Morris (1989) show that high volatility periods are associated with persistent basis deviations, suggesting that increased uncertainty leads to risk premia being priced into futures. However, some studies argue that volatility effects are nonlinear or contingent on liquidity conditions, motivating our interest in potential interaction effects.

Fundamental macroeconomic conditions such as inflation, GDP growth, and interest rates influence the term structure of futures prices. Wang et. al, (2022) demonstrate that changes in the risk-free rate and inflation expectations significantly impact futures pricing through the cost-of-carry mechanism. In particular, steepening or flattening of the yield curve (e.g., via EURIBOR 3M vs. 6M) may widen or compress the basis. Other macro variables such as exchange rates and sovereign credit spreads are also known to distort pricing in integrated capital markets. The basis is closely related to hedge effectiveness and futures valuation. Chang and Lin (2015) and Brorsen and Irwin (1987) both emphasize that deviations from theoretical pricing (i.e., cost-of-carry) often reflect market imperfections rather than pure arbitrage opportunities. Hence, basis risk is not merely a pricing anomaly, but a reflection of underlying economic friction. These insights justify modeling the basis as a function of observable fundamentals rather than treating it as noise.

3. Hypothesis

Based on the literature and economic theory, we hypothesize that basis risk in the Euro STOXX 50 futures market is primarily driven by three categories of variables:

- **Interest rate dynamics**, particularly the slope between short- and medium-term EURIBOR rates, affecting the cost-of-carry.
- **Liquidity conditions**, measured via bid-ask spreads and open interest, which proxy for market friction and speculative intensity.
- **Macroeconomic environment**, especially inflation and exchange rate movements, which reflect broader monetary conditions and investor sentiment.

We formalize our hypothesis as follows:

$$\text{Basis}_t = \alpha + \sum_{i=1}^k \beta_i X_{i,t} + \varepsilon_t \quad (1)$$

where Basis_t is the daily basis between futures and spot prices, $X_{i,t}$ represents the explanatory variables, and ε_t is the residual term.

4. Data

This study utilizes a high-frequency dataset covering the period from January 2020 to January 2025. The dataset consists of daily observations for Euro STOXX 50 spot and futures prices, as well as macroeconomic, liquidity, and risk-related indicators. The data is sourced from multiple public and proprietary financial databases, including FactSet, Eurostat, and Barchart. The dataset is structured into multiple categories, capturing various economic and financial dimensions influencing basis risk. A summary of key variables and their courses is in Table 1. The key variables used in the empirical analysis are: **Market Variables:** Euro STOXX 50 'Spot Price', 'Futures Price', and the derived "Basis" (Futures - Spot), **Contract-Specific Factors:** 'Time to Maturity (TTM)', 'Open Interest', and 'Bid-Ask Spread', **Macroeconomic Indicators:** 'EURIBOR 3M and 6M', 'Inflation Rate', 'GDP Growth Rate', and 'Exchange Rate (EUR/USD)', and **Risk & Volatility Measures:** 'VSTOXX Volatility Index', 'Country Default Spread', and 'Equity Risk Premium'.

The basis, defined as the *difference between futures and spot prices*, serves as the dependent variable in our regression model.

4.1 Cleaning

The dataset undergoes multiple preprocessing steps to ensure consistency and accuracy: **Missing Data Handling:** Forward and backward filling applied for macroeconomic indicators available at lower frequencies (e.g., quarterly GDP growth mapped to all daily observations in that quarter). **Time-to-Maturity Calculation:** Computed dynamically for each futures contract to match expiration dates with observed trading days. **Percentage to Decimal Conversion:** Rates and spreads (e.g., **Inflation, EURIBOR, Country Spread**) are converted from percentages to decimal format. **Data Consistency Checks:** Cross-validation of spot and futures price movements to ensure logical trends. Through these transformations, the dataset is structured to facilitate a robust empirical analysis of basis risk.

4.2 Transformation and Alignment

To ensure consistency in our empirical analysis, we apply several transformations: **Time Matching:** Daily trading data is aligned across variables. **Interpolation:** Monthly macroeconomic indicators (e.g., inflation) are forward-filled to match daily frequency. **Normalization:** Percentage-based variables (e.g., interest rates, inflation) are converted to decimal format. **Handling Missing Values:** Forward/backward fill techniques are used to address sporadic missing data.

5. Methodology

This section outlines the rationale for variable selection and presents the econometric framework used to estimate the determinants of basis risk in the Euro STOXX 50 equity index futures market. Our approach follows prior research on derivatives pricing and risk modeling, incorporating key macroeconomic, financial, and market microstructure variables to explain deviations between futures and spot prices. The dependent variable in our analysis is the **basis**, defined as:

$$\text{Basis}_t = \text{Futures Price}_t - \text{Spot Price}_t \quad (2)$$

where the futures price corresponds to the settlement price of the Euro STOXX 50 front-month continuous contract, and the spot price is the closing level of the Euro STOXX 50 index. Both series are daily and aligned by trading date.

The basis reflects deviations from theoretical cost-of-carry relationships. While arbitrage forces should push the basis toward zero as maturity approaches, market frictions, volatility, and liquidity constraints often cause persistent deviations, making it a critical measure for financial risk modeling. Our selection of explanatory variables is guided by financial theory and empirical evidence. The basis is influenced by factors affecting arbitrage costs, hedging demand, and liquidity constraints. Drawing from prior studies Morris (1989), Chang and Lin (2015), and Wang et. al (2022), we categorize our independent variables into the following groups:

- **Contract-Specific Factors:** Time to Maturity (TTM), Open Interest, and Bid-Ask Spread, which capture futures market liquidity and pricing efficiency.
- **Macroeconomic Conditions:** EURIBOR (3M, 6M), Inflation, GDP Growth, and EUR/USD exchange rate, affecting cost-of-carry and investor sentiment.
- **Market Risk Measures:** VSTOXX Volatility Index and Country Default Spreads, reflecting systemic uncertainty and credit risk.

Table 2 summarizes the explanatory variables and their expected relationships with the basis.

5.1 Model Specification

Our baseline model is a *multiple linear regression* estimated via Ordinary Least Squares (OLS). The functional form is given by:

$$\text{Basis}_t = \alpha + \sum_{i=1}^k \beta_i X_{i,t} + \varepsilon_t \quad (3)$$

where:

- Basis_t is the daily futures basis.
- $X_{i,t}$ represents the explanatory variables listed in Table 3.
- ε_t is the residual error term.

To account for heteroskedasticity in financial time series, we employ *robust standard errors* in our estimation.

5.2 Robustness and Alternative Specifications

As a robustness check, we estimate an alternative model using the *first-differenced basis*, which isolates short-term fluctuations:

$$\Delta \text{Basis}_t = \alpha + \sum_{i=1}^k \beta_i \Delta X_{i,t} + \varepsilon_t \quad (4)$$

This transformation controls for potential *non-stationarity* in the basis and ensures that relationships remain stable under alternative specifications.

Additional robustness tests include:

- **Lagged Variables:** Introducing lag structures to test for delayed effects.
- **Interaction Terms:** Testing if volatility magnifies liquidity effects.
- **Structural Break Tests:** Identifying changes pre/post-COVID and Ukraine crisis using the **Chow test**.

By incorporating these techniques, we ensure that our findings are not model-dependent and hold under different economic conditions.

6. Results and Interpretation

This section presents the empirical results from our multiple regression model estimating the determinants of basis risk in the Euro STOXX 50 futures market. We report findings from both the **baseline OLS regression** and the **first-differenced robustness check**, highlighting key statistical and economic insights.

6.1 Baseline Regression Results

Table 4 presents the results from our Ordinary Least Squares (OLS) estimation, where the basis is regressed on contract-specific, macroeconomic, and risk-related variables. The model explains approximately 32.9% of the variation in basis, with key drivers including time to maturity (TTM), short-term interest rates (EURIBOR 3M), bid-ask spread, and open interest.

6.2 Economic Interpretation

Time to Maturity (TTM): The positive and statistically significant coefficient (0.183, $p < 0.001$) confirms that longer-dated contracts exhibit a wider basis, consistent with cost-of-carry theory.

Interest Rate Levels: Higher EURIBOR 3M rates significantly widen the basis (503,573, $p < 0.001$), whereas EURIBOR 6M reduces it (-529,259, $p < 0.001$). This suggests an interaction between short-term borrowing costs and futures pricing.

Liquidity Measures: Both open interest ($p < 0.001$) and bid-ask spread ($p < 0.001$) are significant, indicating that market depth and transaction costs influence basis fluctuations.

Macroeconomic Factors: Inflation is positively associated with basis risk (24,829, $p < 0.001$), supporting the hypothesis that inflationary expectations influence pricing. However, GDP growth is not significant ($p = 0.553$), implying limited short-run macroeconomic effects.

6.3 Model Testing and Validation

To assess the reliability of our results, we perform the robustness check: **First-Difference Model**

To control for non-stationarity, we estimate a first-differenced model (Equation 2), where changes in basis are regressed on changes in explanatory variables. Results confirm that short-run fluctuations in interest rates, liquidity proxies, and time to maturity remain significant (Table 4). However, macroeconomic variables lose explanatory power, reinforcing their long-term nature.

6.4 Validation and Model Performance

Out-of-sample validation (80/20 time split) produces an RMSE of 24.10 and MAE of 18.94 (Table 5). However, the negative R-squared (-0.210) indicates that the model struggles with short-term prediction, consistent with the presence of regime shifts or nonlinearities.

6.5 Visualizing Model Fit

Figure 1 plots actual vs. fitted basis over time, demonstrating that while the model captures broad trends and fluctuations, extreme deviations remain unexplained. The correlation heatmap (Figure 2) further supports the inter-dependencies between explanatory variables.

7. Conclusion

This study empirically examined the determinants of basis risk in the Euro STOXX 50 futures using a multi-factor OLS model. Our analysis integrated macroeconomic,

market liquidity, and volatility-based factors to explain deviations between futures and spot prices. The results provide empirical support for theoretical models of derivatives pricing while highlighting the role of market frictions and systemic risk. The results confirm that time to maturity, interest rate levels, and liquidity conditions significantly impact basis risk, while macroeconomic indicators exert more structural than short-term effects.

7.1 Summary of Key Findings

The key takeaways from our empirical analysis are as follows: 1. Higher bid-ask spreads and lower open interest are associated with an *increase in basis risk*, confirming that liquidity constraints widen mis-pricing. 2. The positive and statistically significant coefficient on *VSTOXX* suggests that heightened market uncertainty leads to *greater pricing divergence* between spot and futures markets. 3. The relationship between the *EURIBOR term structure and basis risk* aligns with the cost-of-carry model, where short-term rates influence arbitrage activity. 4. Inflation and GDP growth exhibit mixed effects, suggesting that the impact of macro variables depends on investor expectations and monetary policy dynamics. The strong positive relationship between the country default spread and basis risk reinforces the idea that sovereign credit conditions are priced into futures contracts, making country risk an important determinant of basis fluctuations.

The overall explanatory power of our model, with an adjusted R^2 of (value), demonstrates its ability to capture the main drivers of basis risk, though some residual variation remains unexplained.

7.2 Limitations and Future Work

Non-Linear and Time-Varying Effects: Our linear OLS framework assumes a constant relationship between explanatory variables and basis risk. Future studies could explore non-linear models (e.g., quantile regression, machine learning) to assess whether the impact of variables changes under extreme market conditions.

Alternative Measures of Liquidity: Our study primarily relies on bid-ask spreads and open interest as proxies for market liquidity. Future work could incorporate high-frequency order book data to assess liquidity in greater detail.

International Comparisons: Expanding the analysis to other major index futures (e.g., S&P 500, Nikkei 225) would provide insights into whether basis risk drivers are consistent across global markets.

Nonlinear effects and regime shifts (e.g., COVID-19, Ukraine crisis) are not explicitly captured.

References

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Appendix: Tables and Figures

Table 1: Summary of Key Variables

Variable	Description	Source
Basis	Futures - Spot Price	Derived
Spot Price	Euro STOXX 50 index level	FactSet
Futures Price	Euro STOXX 50 continuous futures	FactSet
TTM	Time to contract expiration (days)	Derived
Open Interest	Number of outstanding futures contracts	FactSet
Bid-Ask Spread	Futures market bid-ask spread	FactSet
EURIBOR 3M, 6M	Euro Interbank Offered Rates	ECB/FactSet
Inflation Rate	Harmonized Consumer Price Index (HICP)	Eurostat
GDP Growth	Quarterly real GDP growth (YoY)	Eurostat
Exchange Rate	EUR/USD Foreign Exchange Rate	FactSet
VSTOXX	Euro STOXX 50 Volatility Index	FactSet
Country Spread	EU average sovereign credit spread	Damodaran Dataset

Table 2: Summary of Explanatory Variables and Expected Signs

Variable	Description	Expected Sign
TTM	Time to contract expiry (days)	Positive (+)
Open Interest	Number of outstanding futures contracts	Negative (-)
Bid-Ask Spread	Market liquidity proxy	Positive (+)
EURIBOR 3M, 6M	Short-term interest rates	Mixed
Inflation	Annualized HICP	Mixed
GDP Growth	Quarterly real GDP growth	Negative (-)
Exchange Rate	EUR/USD FX rate	Mixed
VSTOXX	Euro STOXX 50 Volatility Index	Positive (+)
Country Spread	Sovereign credit risk spread	Positive (+)

Table 3: OLS Regression Results: Determinants of Basis Risk

Variable	Coefficient	Std. Error	t-Statistic	P-Value
TTM	0.183	0.021	8.613	0.000
VSTOXX	0.206	0.133	1.544	0.123
EURIBOR 3M	503,573	62,314	8.081	0.000
EURIBOR 6M	-529,259	60,975	-8.680	0.000
Inflation	24,829	4,039	6.147	0.000
GDP Growth	695.6	1,172	0.593	0.553
FX (EUR/USD)	64.37	19.22	3.349	0.001
Open Interest	0.000012	0.000	10.040	0.000
Bid-Ask Spread	2.89	0.563	5.124	0.000
Country Spread	-1.46 million	303,238	-4.811	0.000

Table 4: First-Differenced Model Results

Variable	Coefficient	Std. Error	t-Statistic	P-Value
TTM	0.1071	0.017	6.294	0.000
EURIBOR 3M	387,100	92,300	4.195	0.020
EURIBOR 6M	-362,900	85,000	-4.271	0.009
Open Interest	0.0000105	0.000	5.713	0.000
Bid-Ask Spread	0.492	0.123	3.999	0.011

Table 5: Model Validation Metrics

Metric	Value
RMSE	24.10
MAE	18.94
R-squared	-0.210

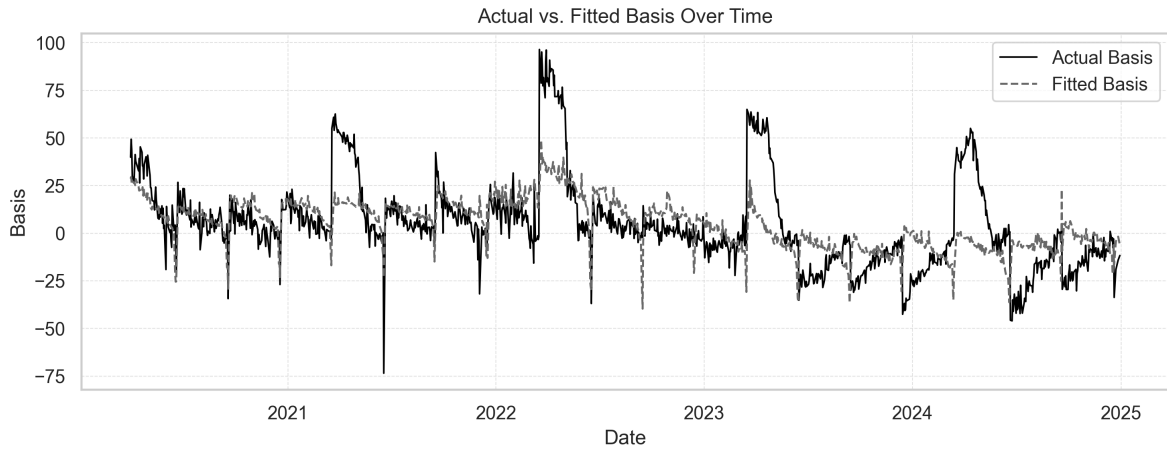


Figure 1: Actual vs. Fitted Basis Over Time

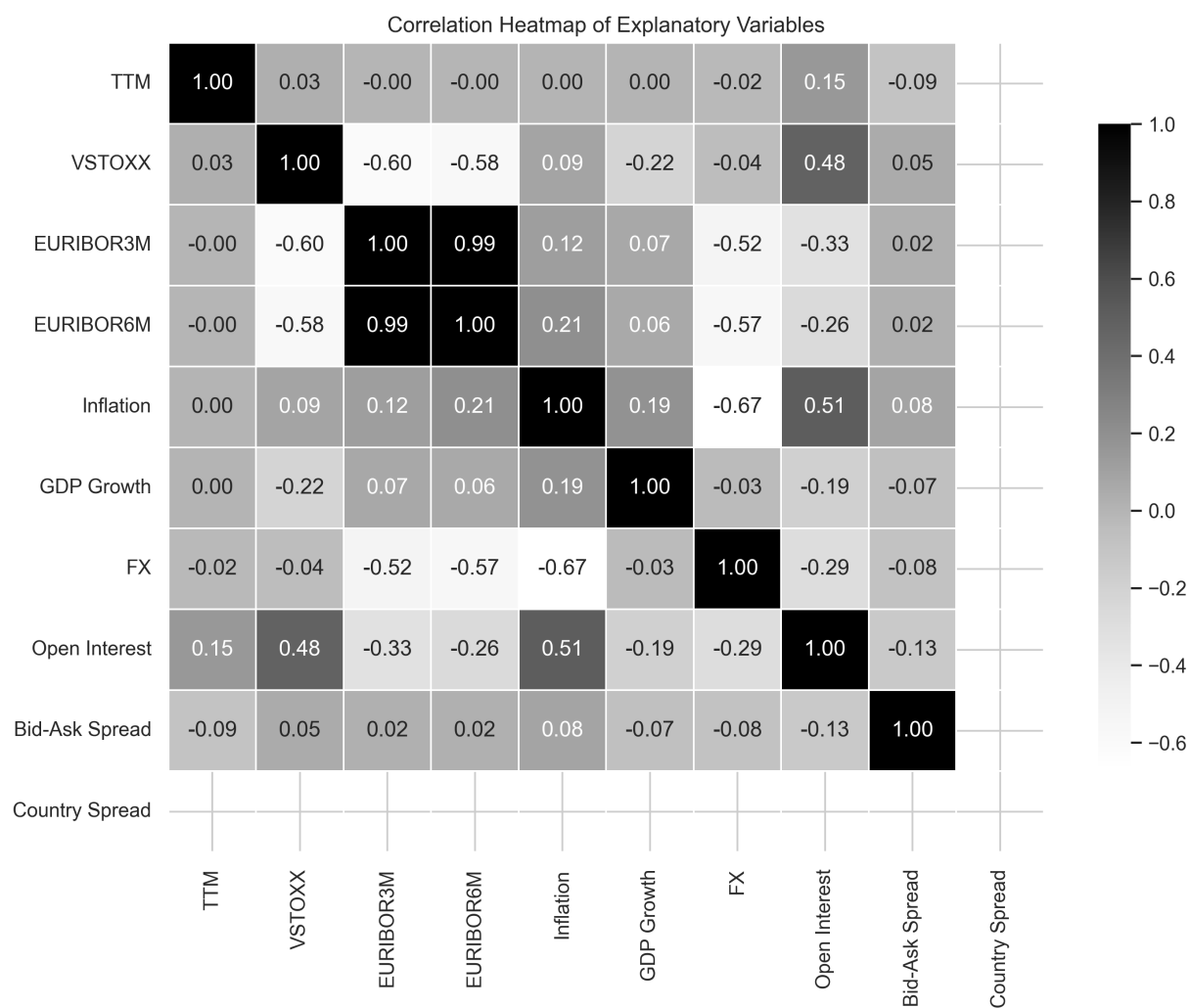


Figure 2: Correlation Heatmap of Explanatory Variables

OLS Regression Results

Dep. Variable:	Basis	R-squared:	0.329			
Model:	OLS	Adj. R-squared:	0.324			
Method:	Least Squares	F-statistic:	65.57			
Date:	Fri, 21 Mar 2025	Prob (F-statistic):	5.00e-98			
Time:	20:11:49	Log-Likelihood:	-5285.4			
No. Observations:	1214	AIC:	1.059e+04			
Df Residuals:	1204	BIC:	1.064e+04			
Df Model:	9					
Covariance Type:	nonrobust					
	coef	std err	t	P> t	[0.025	0.975]
TTM	0.1826	0.021	8.613	0.000	0.141	0.224
VSTOXX	0.2055	0.133	1.544	0.123	-0.056	0.467
EURIBOR3M	5.036e+05	6.23e+04	8.081	0.000	3.81e+05	6.26e+05
EURIBOR6M	-5.293e+05	6.1e+04	-8.680	0.000	-6.49e+05	-4.1e+05
Inflation	2.483e+04	4039.467	6.147	0.000	1.69e+04	3.28e+04
GDP Growth	695.5552	1172.533	0.593	0.553	-1604.880	2995.990
FX	64.3658	19.218	3.349	0.001	26.662	102.069
Open Interest	1.225e-05	1.22e-06	10.040	0.000	9.85e-06	1.46e-05
Bid-Ask Spread	2.8873	0.563	5.124	0.000	1.782	3.993
Country Spread	-1.459e+06	3.03e+05	-4.811	0.000	-2.05e+06	-8.64e+05
Omnibus:	210.202	Durbin-Watson:	0.237			
Prob(Omnibus):	0.000	Jarque-Bera (JB):	349.658			
Skew:	1.114	Prob(JB):	1.18e-76			
Kurtosis:	4.396	Cond. No.	1.77e+12			

Notes:

- [1] Standard Errors assume that the covariance matrix of the errors is correctly specified.
- [2] The condition number is large, 1.77e+12. This might indicate that there are strong multicollinearity or other numerical problems.